

Chapter 15: 5G/6G-Enabled Edge Intelligence

Arun Kumar Rathore¹, Sachin Lakhera², Shweta Gupta³

^{1,2,3}J.B. Institute of Technology, Dehradun, Uttarakhand

¹arunkumarrathore706@gmail.com

²24shwtgupta87@gmail.com

Abstract

A paradigm shift toward edge intelligence, where data processing, analytics, and AI inference are carried out closer to the end users and sources, is made possible by the quick development of 5G and the next 6G wireless networks. To meet the demanding needs of next-generation applications like autonomous vehicles, smart cities, industrial automation, extended reality, and tactile internet, 5G/6G-enabled edge intelligence combines ultra-low-latency communication, high data rates, massive connectivity, and native AI support. By implementing intelligent models at the network edge, this paradigm lowers backhaul traffic, boosts privacy, increases dependability, and ultimately permits real-time decision-making. Several technologies, including cloud-native architectures, network slicing, and multi-access edge computing, enable edge intelligence in 5G networks. By incorporating AI-native air interfaces, semantic communications, integrated sensing and communication, and distributed learning frameworks like federated and split learning, 6G is anticipated to further develop edge intelligence in the future. There are still a number of obstacles to overcome in this prospective environment, including resource-constrained edge devices, changing network conditions, model management, security, and energy efficiency. The architectural underpinnings, enabling technologies, use cases, and unresolved research issues for 5G/6G-enabled edge intelligence are highlighted in this abstract. In order to realize intelligent, scalable, and sustainable future networks, this handout emphasizes how the tight coupling of improved wireless communications with AI-driven edge computing will be crucial.

Keywords: Artificial Intelligence, Ultra-Low Latency, Federated Learning, 5G, 6G, Edge Intelligence, Integrated sensing and Communication; Sustainable future Networks

1. Introduction

The launch of 5G technology has begun to occur through business services, and as 6G technology emerges, a lot of researchers are working on ways to make it happen. With their larger system capacity, higher speeds in transmitting information, very low levels of delay, and very little power used, both 5G and 6G will help create an even better Internet of Things (IoT) than exists today. In addition, with the advances being made in edge computing, virtual networks, and artificial intelligence (AI), there is also a lot of possibility for the Full IoT (FIoT) to be created [1]. Overall, AI and 5G/6G are thought to play a significant role in creating the complete IoT by resolving the following challenges:

- a. How AI can help provide flexible spectrum usage in 5G/higher frequency 6Gs.
- b. New management techniques for networks to provide for the effective coexistence of enhanced mobile broadband (eMBB) and IoT services, such as software-defined networks (SDN), Network Functions Virtualization (NFV), and dynamic network slicing, etc.

- c. The many various security and privacy concerns associated with the FIoT, including out-of-date firmware, inadequate passwords/credentials, malware/ransomware, expectations/prevention of attacks, privacy of data, etc.
- d. The health care industry is extremely complex. To begin, a health system is made up of various components that are connected. WHO describes a health system as all the institutions, individuals, and activities focused on preventing, improving, or restoring one's health.

Health systems can be defined broadly, and may contain both rural/urban areas, and contain both more formal/allopathic care systems and less formal/traditional care systems; and are mostly defined as national health institutions. As part of his/her job, those working within a health system should identify community health challenges, identify unmet needs, assess current available resources, develop SMART (Specific, Measurable, Achievable, Relevant, Time-limited) goals to help resolve those needs, and establish an administrative structure to successfully meet these goals. To enhance their efficiency in carrying out their mission, health systems should consider the incorporation of risk management, performance management, and management by objectives principles. The general public must be made aware of the current state of their health and how to improve that state. In addition, as new diseases emerge and shape how healthcare policy should respond, the necessary data to create knowledge and intelligence must be utilized [2]. These innovative shifts in health and illness will be complemented by historical successes and present trends in both the medical and social aspects of society. Becoming self-sufficient will allow for decreased overall health care costs for health systems to operate. With increased access to high-quality, easily accessible, and user-friendly information, the ability to self-direct medical care will become paramount for controlling disease. In a manner analogous to that of meteorological prediction, thorough planetary and ecological mapping could provide for the prediction of disease outbreaks. Interdisciplinary health care workers will meet the needs of the workforce shortage. A health information system (HIS) would significantly improve both the quality and efficiency of health-related services in many different settings [3]. However, technology-dependent systems are just as susceptible to social, historical, and geographic conditions, which will affect both the basic technology components and the level of importance and function of each technology component, and the sequence in which they develop [4]. Through increasing productivity and South America-wide information sharing, e-health solutions expand the capacity for human resource use among communities and individual workers. Individuals with knowledge or skills should receive lifelong training by suitably trained professionals and continue to have access to ongoing external evaluation against established indicators. In addition, both medical and non-medical fields should have limited tracking of environmental changes and barriers to improvement as an integral component of their efforts [5]. EMR systems facilitate health services to users in a single organization by recording and monitoring consumer-related transactions, and are effective at promoting high-quality medical care to that user. Most modern health information technology uses some form of electronic health record (EHR) to prepare, store, and retrieve your health records. They will use your official medical records and develop an electronic version of those records for their establishment or group. For example, an EHR will include all patient data such as laboratory results, vital signs, provider notes, immunization records, use of any medications, and past health history [6].

2. Healthcare's Edge Computing: Edge

Computing growth has increased with the rise of connected devices in healthcare and the need for real-time data analysis. Edge Computing is defined as the processing of data close to the point it is generated using computation resources at the edge of the Computer Network. This leads to faster response times and less latency in transmitting data to the central processing facility. 5G networks provide the critical infrastructure for Healthcare Edge Computing, allowing faster processing and quicker decisions. Edge Computing is the backbone of Healthcare AI and ML. AI and ML have the potential to transform Medicine, specifically in the areas of reading images and developing personalized treatment plans. AI algorithms can be implemented at the edge of the 5G & 6G networks due to the rapid speed & low latency that accompany these new technologies. Medical data will be processed in real time as a result of this implementation. As a result, many researchers are exploring the potential of combining both Artificial Intelligence (AI) & Machine Learning (ML) with both 5G & 6G technologies to develop more intelligent healthcare systems with enhanced predictive capabilities and decision support systems. Utilization of VR/AR in healthcare [7-10] is also rapidly growing in applications such as surgical planning, medical education, pain control, & therapeutic practices. There is sufficient bandwidth & low enough latency in order to provide realistic augmented/virtual reality experiences through 5G networks, while the use of 6G technologies in the healthcare industry is being studied to enhance the quality, responsiveness, & interactivity of augmented/virtual reality applications. Regulatory issues and standards will also need to be taken into consideration with regard to these new technologies as they relate to the use of AR/VR within healthcare [11-14]. To create standards and guidelines for using new technologies such as 5th generation (5 G) and 6th generation (6 G) networks in the field of health care, regulatory agencies and standard-setting bodies are putting forth a lot of effort. These new technologies will save money and improve patient care, but will be focused on providing health care. The initiatives represent the three main focuses: interoperability between different types of devices, compliance with privacy regulations, and quality of service. The goal is to raise the standard for delivering health care.

3. Edge Computing in IoT

In the last few years, there has been a rise in the number of solutions that decentralize communication, data collection, and processing from the data center to the edge and then migrate all these capabilities to the edge [15]. This development has resulted in the emergence of Edge Computing, which refers to the computing processes located at the network edges as opposed to through the cloud (i.e., at the nearest point to the data source), in order to carry out computations as close as possible to the network edges [16]. Thus, by eliminating or reducing latency while also reducing cost and increasing responsiveness, scalability, dependability, security, and privacy, Edge Computing helps computer systems to work more effectively than traditional computing systems [17]. The Internet of Things (IoT) is also an established standard for systems that process and collect data and has become a major area of study and technology [18]. Internet of Things systems consist of many heterogeneous devices (sensors, vehicles, machines, household appliances, and meters) that communicate over multiple different types of communication protocols [19]. IoT solutions can provide benefits across a range of different fields, such as the following: Industry 4.0, smart homes, smart cities, and healthcare [20-23]. The wide range of resources associated with IoT networks creates challenges in terms of managing these resources, resulting in problems related to communication protocol compatibility, real-time operations, data management, large volumes of data

storage, security, and privacy. Edge Computing architectures provide a way to support the varying forms of data generated by IoT devices by supporting the overall infrastructure of the IoT in this context [24, 25]. A simple three-level Edge Computing architecture is illustrated in Figure 1.

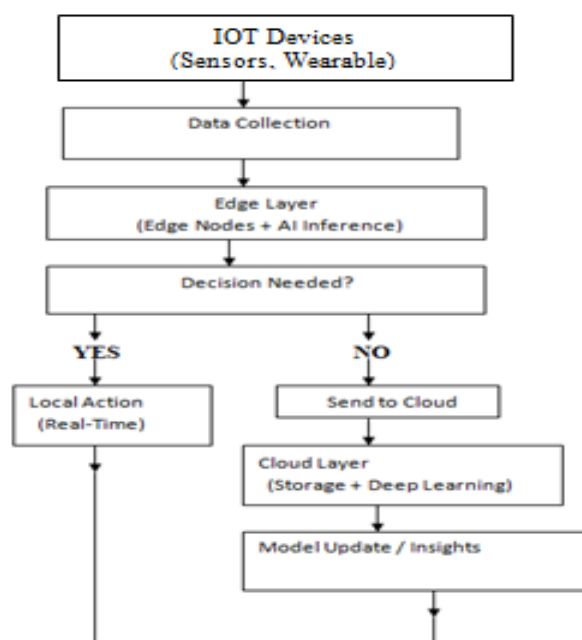


Figure 1. Flowchart of 5G/6G-enabled edge intelligence illustrating data processing across IoT devices, edge nodes, and cloud infrastructure [26].

Real-time decisions are performed at the edge, while complex analytics and model training are executed in the cloud, with continuous feedback to improve edge intelligence. There are three levels for computer architecture as follows:

- Level 1: IoT + Sensors:** This level (which is the bottom level of the architecture) consists of a combination of people and IoT devices (e.g., smart plugs, smart electricity meters, and various sensors). Each of these devices will be ingesting data and performing actions associated with that data at this level.
- Level 2: Edge Nodes:** Edge nodes (which are the second level) are the devices primarily responsible for performing computing/routing/processing of data contained in each of the Edge devices.
- Level 3: Cloud Services:** The third level consists of cloud services that have a much greater processing requirement. The services performed by this level include analytics, artificial intelligence, machine learning, and visualization of data.

4. The influence of 5G on Edge Intelligence

In order to facilitate real-time communication, local computing, very high speed data transmission, and very high availability, the majority of use cases for 5G are reliant upon Edge Computing; these include but are not limited to: healthcare; remote surgery, diagnostics, and monitoring data and Vital Signs, where doctors can use a remote surgery operation platform to perform operations via the

Internet on patients that are far away from them (i.e., that the doctor feels that their patient is safe and secure and so therefore is now able to control the surgical tools to perform the operations on their patients from a distance); multimedia and entertainment usages; Examples of multimedia and entertainment usages are streaming capability for 3D & HDTV, and immersive Augmented, Mixed or Virtual Reality experiences, that require the capability to stream video content into VR glasses [27]. Moving the computation load from the glasses (of the user) to Edge Servers will also allow users to have smaller & lighter VR glasses. The next evolution of the Internet of Things, which introduces Internet of Tactile (IoT), is expected to provide very low-latency, high-reliability network connections that will ensure real-time delivered control messages and real tangible tactile experiences across relatively long distances in a secure manner. Using URLLC allows for a low-latency transmission of small amounts of data with a high level of reliability [28]. For example, when a fire alarm goes off, URLLC provides a high level of reliability in M2M communications between various UE devices. The IoT (Internet of Things) connects everyday devices (like kitchen appliances) to the internet. Future manufacturing plants, such as smart factories, will automate & streamline operations through robotics [29]. Through an intelligent transportation system (ITS), drivers can provide/receive information through traffic management centers to avoid other vehicles that could pose a risk of collision by providing real-time updates on sudden stops due to accidents. Unmanned ground vehicles will be able to sense their environment around them and navigate independently through the use of sensors, cameras & other technology.

5. The development of Sixth Generation (6G)

Technologies began in 2019 and 2020 with new high-performance requirements for the anticipated evolution of Internet Applications; specifically, transitioning from the Internet-of-everything to An Intelligent Internet-of-Everything will entail higher-speed performance, improved energy efficiency, lower but larger numbers of connections that can be controlled responsively and reliably, connected intelligent devices that provide machine learning and deep learning capabilities, and the ability to use a wider range of frequency bands for wireless communications [30]. Three types of new applications will support 6G technologies: Computation-oriented communications (COC), contextually Adaptable explainable mobile broadband (CAeC), and Event/data-orientated Ultra-reliable low-latency communication (EDURLLC). COC will use a selection of resources (i.e., the rate and latency performance of the connection) from the available (rate, latency, reliability) resource space to meet the required computational accuracy for machine-learning algorithms [31]. CAeC will provide eMBB cellular services that can adapt dynamically to network congestion levels, network traffic patterns, changes in network topologies, user mobility, and user behavior within a social network [32].

6. AI-Powered New Networks for 6G

The disruptive growth of smart city mobile apps and associated demand from providers to meet strict service level agreements (SLAs) is pushing the development of 6G. These new applications depend on 6G, along with AI and big data analytics, to be able to be premium due to their high quality of service (QoS) requirements. The capabilities that will be offered by 6G will be much greater than any features found in 5G LTE. They will include battery-less devices that are extremely high-speed, low-latency, and offer extreme energy savings; wide bandwidths and extensive global, broadband coverage [33]. In addition to the aforementioned features, 6G will also need context-aware algorithms to better optimize the architecture, protocols, and operations of the new network. Through

the incorporation of linked intelligence into its overall design, 6G networks will seamlessly blend the communication, computing, and storage capabilities of both edge devices and the cloud and core network, resulting in the highest possible level of performance [34]. An optimized, AI-enabled 6G network will facilitate the delivery of low-latency, highly reliable, secure, and fast execution across a wide range of applications. Traditional methods of delivering an optimal experience will be challenged by the length of time (time) between the analysis and the generation of a response. Traditional methods of performing statistical analysis (e.g., using past experience, knowledge) to provide data for decision-making are unlikely to produce a successful outcome when there is a large time lapse between the time the analysis is completed and the time a decision is made. Thus, Machine Learning (ML) and Deep Learning (DL) algorithms are being used to find solutions to many of the issues associated with networking, including data offload/caching [35]. This section provides a description of the AI-based protocols and processes used in 6G networks (depicted in Figure 2) as well as the ML/DL self-learning models used in these networks.

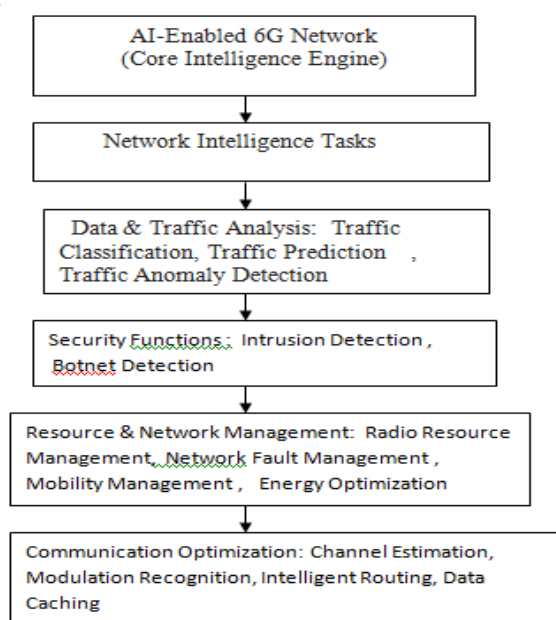


Figure 2: Flowchart representation of artificial intelligence-enabled 6G networks illustrating key functional modules, including traffic analysis, security management, resource optimization, and communication enhancement tasks [36].

7. Taxonomies of Next Generation 5G and 6G Technologies

6G, the wireless technology of the future, is expected to be commercially available as early as late 2020 or early 2030. By 2030, 6G should have peak data transfer rates up to (1) terabits per second (Tbps), extreme (2) low latencies (approximately 1 millisecond), and will provide new healthcare applications and services that could not take advantage of prior generations of wireless technologies. New applications and services in healthcare that could not have been offered by earlier (3) generations of wireless technologies are also anticipated to be available with 6G. 6G technology networks will, for example, be able to support ultra-high-definition medical imaging and remote surgeries [37]. However, some experts have expressed concerns about potential risks and challenges

associated with implementing 6G in the healthcare industry. A number of key goals are provided in Table 1 for enhancing both 5G and 6G networks. Classification of 6G-enabled edge intelligence applications across core technologies, communication paradigms, and IoT domains, highlighting their role in healthcare, transportation, industry, and smart environments.

Table 1: Classification of 6G-enabled edge intelligence applications [38-39].

Category	Domain / Technology	Applications / Use Cases
Core Technology	Artificial Intelligence	Smart connected healthcare, intelligent transportation, energy trading, digital certificates, fraud detection
Core Technology	Block chain	Secure transactions, digital identity, real estate management, rights management
Core Technology	Edge-Cloud Computing	Disease diagnosis, content delivery, data analytics, content caching
Core Technology	Extended Reality (XR)	3D medical imaging, immersive training, remote repair, virtual tourism
Communication	Holographic Communication (HC)	Immersive gaming, 3D mapping, remote healthcare, robotics control
IoT Domain	Internet of Medical Things (IoMT)	Remote patient monitoring, disease diagnosis, smart drug management, remote surgery
IoT Domain	Internet of Vehicles (IoV)	Smart parking, accident prevention, collision warning, autonomous driving
IoT Domain	Internet of Robotic Things (IoRT)	Crop harvesting, medicine delivery, warehouse automation, robotic assembly
IoT Domain	Industrial IoT (IIoT)	Predictive maintenance, quality control, safety management, supply chain optimization
IoT Domain	Internet of Drones (IoD)	Surveillance, disaster management, traffic control, live event streaming
System Layer	Internet of Things (IoT)	Integration platform connecting devices, sensors, and smart applications

8. Comparison of 5G and 6G Technologies

6G is currently being researched and developed; however, once developed, this technology will enable substantial advancements in health care through faster, more dependable and more secure communication between patients and medical practitioners. Additionally, the mobility capabilities of

5G technology differ from those of previous generations of wireless technology in terms of speed (i.e., 5G develops as a complete upgrade from 4G); however, there are several use cases that were designed through the design process of this system. Due to the factors previously mentioned, how does 5G outperform all previous generations of wireless networks? There are three primary areas where the 5G network offers improvements [40].

- I. **Reliability/Security** – Reliability claims (99.999% reliable) for the 5G network; therefore, 5G provides extremely reliable and secure networks. Reliability and security can be achieved through various technological advances, including data-in-motion encryption and Network Slice Technology (NST) [41] and/ or reliable handoff from radio access to the network. Because the infrastructure of the 5G network is evolving toward virtualized (virtual machine) and software-defined (defined by software) networks, a potentially large decline in network incidents (disruption or outage) is anticipated as this transition occurs. All forms of previous generation connectivity (4G, Wi-Fi, Bluetooth, etc.) do not provide this type of reliability/security.
- II. **Performance:** The implementation of 5G will dramatically improve the overall performance of networks and the use of mobile services. It has been shown that 5G technology will have 90% lower latency when compared to 4G and LTE5 technologies. Additionally, the maximum data transmission rate with 5G has also been shown to support up to 20 GB/s of data when compared to a maximum of 10 MB/s with 4G and LTE technology.
- III. **Capacity:** Projections indicate that by 2026, there should be approximately 29 billion devices connected to the Internet throughout the world. This represents a significant increase from the currently estimated number of connected devices around the world. The current connectivity technologies, such as Bluetooth, Zigbee, Low Power Wide Area Networks (LPWAN), and Narrowband Internet of Things (NB-IoT), will struggle to support this new level of device density. In reality, 5G networks will have the capacity to support at least 1 million connected devices through the use of 5 Gbps wireless connections.
- IV. **6G Horizon:** In the future of smart healthcare, the primary delivery of healthcare services will be location-independent and customized to each patient's unique needs. Advancements in mobile communications infrastructure will ensure that new applications can be developed and implemented. Such applications include real-time monitoring of an individual's health status, diagnosing and determining the cause of disease or injury remotely, providing holographic training for medical procedure preparation/recovery, and delivering remote surgical care. The improvements to 6G's sensing and artificial intelligence capabilities will help provide a host of options for real-time analysis of medical data [42].

9. The implementation of 5G/6G Edge

Intelligence technologies have had a drastic impact on the modern-day digital ecosystem by allowing for distributed intelligence, ultra-low-latency communication, and real-time processing of data. These characteristics of edge intelligence, when combined with 5G technologies currently available and upcoming 6G technologies, will help to enable the large-scale utilization of data, as well as the ability to support applications in industries such as transportation, healthcare, and smart cities where high levels of data and low-latency communication are required.

10. Intelligent Healthcare Systems & Autonomous Vehicles

In the healthcare arena, edge intelligence enabled by 5G/6G systems will allow for intelligent diagnostics, real-time patient monitoring, and more. The data required to perform these functions is generated continuously via wearable devices and IoT-enabled devices used in patient care environments (e.g., wearables, IoT medical devices, etc.), and will need to be processed at the edge (e.g., at the nearest edge node) for fast analysis. The low latency and high reliability offered by using 5G/6G for these types of applications will drastically improve existing applications and enable new applications for robotic surgery, AI-assisted diagnoses, remote patient monitoring, and smart ambulance technologies, while also improving data privacy and security through processing sensitive medical data at the edge [43]. Autonomous vehicles process data via sensors in real time so that they have a safe method of travel, and edge intelligence allows for communication between vehicles and their surroundings through vehicle-to-everything (V2X) communication. The ultra-low latency provided by the 5G/6G networks meets the requirements for autonomous vehicles' decision-making processes, predictions of traffic, and avoidance of collisions [44].

11. Smart City & Automation in Industry (Industry 4.0)

Smart city applications utilize the edge intelligence of many of the urban processes that occur to be managed successfully. Real-time analytics enable intelligent traffic systems, smart surveillance, energy management, and environmental monitoring. 5G/6G networks provide high-speed connections with high capacity and many IoT solutions installed in a connected city [45]. The use of edge intelligence allows for the monitoring and controlling of actual manufacturing process data in order to improve automation and operational efficiency within an industrial environment. This enhanced automation leads to improved production capabilities and decreased downtime due to reduced dependence on the cloud infrastructure.

Augmented and virtual reality (AR/VR)

The use of augmented and virtual reality (AR/VR) apps requires a very low latency rate and high data throughput in order for users to have an immersive experience. The real-time rendering and processing of AR/VR content is accomplished through edge intelligence, and 6G networks will improve the capabilities of these applications even further with their support of holographic communications and ultra-high-definition streaming.

Intelligent Farming & Disaster Management

In precision agriculture, edge intelligence provides the ability to monitor soil conditions, crop health, and other environmental indicators in real-time. Applications that provide increased agricultural productivity and greater resource efficiency through automated farming systems, smart irrigation, and insect detection using edge intelligence [46]. In catastrophic management, edge intelligence allows for the rapid processing of data and immediate decision-making capabilities during an emergency situation. The use of low-latency communication coupled with distributed intelligence

will lead to significant reductions in response times for applications such as emergency response coordination, drone-based monitoring, and early warning systems.

12. Challenges

As a result of the overall increase in data generation, collection, and use, data security and privacy are now paramount considerations for 5G and future 6G networks. Adversarial AI models could potentially degrade the link for communications and subsequently affect network performance, while extracting sensitive data can result in significant consequences operationally and financially [47]. In traditional networks, decision-making is primarily handled by human operators, and in AI networks, it is moving toward automation. Human involvement is mainly due to extraordinary events, serving as a mechanism for safety. This change raises concerns about responsibility, trust, and the resiliency of the system. Due to the rapid growth of mobile users and connected devices, the complexity of networks has grown exponentially. Quality of Service (QoS) remains a major challenge for network operators when managing a multitude of different types of devices. Therefore, network management capabilities that are scalable and efficient are essential [48]. Another challenge is energy efficiency. Energy-efficient devices can reduce overall energy consumption and improve transmission range. Additionally, because of the extensive use of small cells in various network architectures, resource allocation and interference management become increasingly difficult. Additionally, effective coordination of processing, storage, and communication resources is required as part of network slicing. The technologies described include mobile edge computing (MEC), edge caching (EC), distributed data store (DDS), and digital signal processing (DSP) devices. It is assumed that complex artificial intelligence models will assist radio access networks (RAN) with adapting to changing service requirements; however, it is still a challenge to implement such intelligent systems in practice.

13. Directions for the Future

Emerging areas of new research will have a strong impact on the evolution of AI-based 5G/6G networks. One of the most promising advances in this space is Quantum Artificial Intelligence, which will significantly reduce computational complexity and increase the overall effectiveness of deep-learning models. This will have a huge influence on optimization and decision-making processes in the next-generation communications system [49]. The second of the critical success factors to be aware of when considering the ethical adoption of AI into communications networks relates to the ethical and societal considerations of AI-based decision-making (e.g., Fairness, Transparency, Accountability, and User Privacy). The creation of regulations and cooperation amongst all Stakeholders will play a critical role in the long-term success of this area. Future research will also focus on the integration of AI into other advanced technologies (e.g., Blockchain, Internet of Things - IoT, etc.), resulting in intelligent, decentralized, and secure communication systems. AI-based optimization techniques (e.g., Game Theory) will serve as the basis for modeling user behavior and optimally allocating resources in increasingly complex network environments [50]. Overall, it is anticipated that 5G/6G networks will evolve to become highly autonomous and adaptive systems that will enable intelligent interaction between users, devices, and service providers. By continually learning and optimizing their performance, data-driven, intelligent communication ecosystems are expected to be enabled through the evolution of these networks.

14. Conclusion

Edge intelligence, combined with new cellular communications technologies, represents a transformative shift in how IoT systems are architected. 5G and 6G-enabled edge intelligence can create a high-performance system for processing large volumes of real-time data and making complex economic and social decisions through the integration of AI, edge computing, and high-speed communication networks. This type of paradigm will enable advanced applications within the healthcare context, such as intelligent emergency response systems, AI-assisted diagnosis, and remote patient monitoring. Although a number of technical barriers exist today, continued research and technology development will enhance the performance of these Edge Intelligence Systems in the future. In addition to supporting continued growth of intelligent healthcare systems, the evolution of 6G networks will further increase the confluence of decentralized AI and high-performance communication networks.

References

- [1]. McCoy, D., & Allotey, P. (2021). An introduction to health systems (pp. 3–16). Cambridge University Press.
- [2]. Durrani, H. (2016). Healthcare and healthcare systems: Inspiring progress and future prospects. *mHealth*, 2, Article 3.
- [3]. Walsham, G., & Sahay, S. (2006). Research on information systems in developing countries: Current landscape and future prospects. *Information Technology for Development*, 12(1), 7–24.
- [4]. Broyles, D., Crichton, R., Jolliffe, B., Sæbø, J. I., & Dixon, B. E. (2016). Shared longitudinal health records for clinical and population health. In B. E. Dixon (Ed.), *Health information exchange: Navigating and managing a network of health information systems* (pp. 149–162). Academic Press.
- [5]. El Taguri, A. (2008). Essential concepts in modern health services. *Libyan Journal of Medicine*, 3(3), 1–12.
- [6]. Coiera, E. (2009). Building a national health IT system from the middle out. *Journal of the American Medical Informatics Association*, 16(3), 271–273.
- [7]. Liu, E., Effiok, E., & Hitchcock, J. (2020). Survey on health care applications in 5G networks. *IET Communications*, 14(7), 1073–1080.
- [8]. Ahad, A., Jiangbina, Z., Tahir, M., Shaye, I., Sheikh, M. A., & Rasheed, F. (2024). 6G and intelligent healthcare: Taxonomy, technologies, open issues and future research directions. *Internet of Things*, 25, 101068.
- [9]. Ngiam, K. Y., & Khor, W. (2019). Big data and machine learning algorithms for health-care delivery. *The Lancet Oncology*, 20(5), e262–e273.
- [10]. Toh, C., & Brody, J. P. (2021). Applications of machine learning in healthcare. In *Smart manufacturing: When artificial intelligence meets the Internet of Things* (pp. 65–90). Elsevier.
- [11]. Van der Schaar, M., Alaa, A. M., Floto, A., Gimson, A., Scholtes, S., Wood, A., McKinney, E., Jarrett, D., Lio, P., & Ercole, A. (2021). How artificial intelligence and machine learning can help healthcare systems respond to COVID-19. *Machine Learning*, 110(1), 1–14.
- [12]. Rubinger, L., Gazendam, A., Ekhtiari, S., & Bhandari, M. (2022). Machine learning and artificial intelligence in research and healthcare. *Injury*, 54(Suppl. 1), S69–S73.
- [13]. Moro, C., Stromberga, Z., Raikos, A., & Stirling, A. (2017). The effectiveness of virtual and augmented reality in health sciences and medical anatomy. *Anatomical Sciences Education*, 10(6), 549–559.

- [14].Ferrari, V., Klinker, G., & Cutolo, F. (2019). Augmented reality in healthcare. *Journal of Healthcare Engineering*, 2019, Article ID 9321535.
- [15].Ma, M., Jain, L. C., & Anderson, P. (2014). Future trends of virtual, augmented reality, and games for health. In M. Ma et al. (Eds.), *Virtual, augmented reality and serious games for healthcare* (Vol. 1, pp. 1–6). Springer.
- [16].Bhugaonkar, K., Bhugaonkar, R., & Masne, N. (2022). The trend of metaverse and augmented & virtual reality extending to the healthcare system. *Cureus*, 14(9), e29071.
- [17].St Cyr, T. J. (2013). An overview of healthcare standards. In *Proceedings of IEEE SoutheastCon* (pp. 1–5). IEEE.
- [18].Schulz, S., Stegwee, R., & Chronaki, C. (2019). Standards in healthcare data. In *Fundamentals of Clinical Data Science* (pp. 19–36). Springer.
- [19].Kok, J. (2021). A standard story: On the use and consequences of standards in healthcare regulation (Doctoral dissertation).
- [20].Neuhaus, C., Polze, A., & Chowdhury, M. M. R. (2011). Survey on healthcare IT systems: Standards, regulations and security (Technical Report No. 45). Universitätsverlag Potsdam.
- [21].Kethareswaran, V., & Ram, C. S. (2017). An Indian perspective on the adverse impact of Internet of Things (IoT). *Advances in Distributed Computing and Artificial Intelligence Journal*, 6, 35–40.
- [22].Xu, L. D., Xu, E. L., & Li, L. (2018). Industry 4.0: State of the art and future trends. *International Journal of Production Research*, 56, 2941–2962.
- [23].Sittón-Candanedo, I., Hernández-Nieves, E., Rodríguez-González, S., Santos-Martín, M. T., & González-Briones, A. (2018). Machine learning predictive model for Industry 4.0. In *International Conference on Knowledge Management in Organizations* (pp. 501–510). Springer.
- [24].Garcia, Ó., Alonso, R. S., Prieto, J., & Corchado, J. M. (2017). Energy efficiency in public buildings through context-aware social computing. *Sensors*, 17(4), 826.
- [25].Kaur, N., & Sood, S. K. (2017). An energy-efficient architecture for the Internet of Things. *IEEE Systems Journal*, 11(2), 796–805.
- [26].González-Briones, A., De La Prieta, F., Mohamad, M., Omatu, S., & Corchado, J. M. (2018). Multi-agent systems applications in energy optimization problems: A state-of-the-art review. *Energies*, 11(8), 1928.
- [27].Stojkoska, B. L. R., & Trivodaliev, K. V. (2017). A review of Internet of Things for smart home: Challenges and solutions. *Journal of Cleaner Production*, 140, 1454–1464.
- [28].Chamoso, P., González-Briones, A., Rodríguez, S., & Corchado, J. M. (2018). Tendencies of technologies and platforms in smart cities: A state-of-the-art review. *Wireless Communications and Mobile Computing*, 2018, 3086854.
- [29].Talari, S., Shafie-Khah, M., Siano, P., Loia, V., Tommasetti, A., & Catalão, J. P. S. (2017). A review of smart cities based on the Internet of Things concept. *Energies*, 10(4), 421.
- [30].Dang, L. M., Piran, M. J., Han, D., Min, K., & Moon, H. (2019). A survey on Internet of Things and cloud computing for healthcare. *Electronics*, 8(7), 768.
- [31].Lin, J., Yu, W., Zhang, N., Yang, X., Zhang, H., & Zhao, W. (2017). A survey on Internet of Things: Architecture, enabling technologies, security and privacy, and applications. *IEEE Internet of Things Journal*, 4(5), 1125–1142.
- [32].Yu, W., Liang, F., He, X., Hatcher, W. G., Lu, C., Lin, J., & Yang, X. (2018). A survey on edge computing for the Internet of Things. *IEEE Access*, 6, 6900–6919.
- [33].Aazam, M., Harras, K. A., & Zeadally, S. (2019). Fog computing for 5G tactile industrial Internet of Things: QoE-aware resource allocation model. *IEEE Transactions on Industrial Informatics*, 15(5), 3085–3092.
- [34].Rao, J., & Vrzic, S. (2018). Packet duplication for URLLC in 5G: Architectural enhancements and performance analysis. *IEEE Network*, 32(2), 32–40.

-
- [35].Letaief, K. B., Chen, W., Shi, Y., Zhang, J., & Zhang, Y. J. A. (2019). The roadmap to 6G: AI-empowered wireless networks. *IEEE Communications Magazine*, 57(8), 84–90.
- [36].Ismail, L., Hagimont, D., & Mossiere, J. (2000). Evaluation of mobile agents technology. *Information Science and Technology*, 19, 1–17.
- [37].Ismail, L., & Belkhouche, B. (2009). Mobility management for mobile agents. In *Future Internet Conference* (pp. 31–38).
- [38].Hagimont, D., & Ismail, L. (2000). Mobile agents vs. client/server performance comparison. *Techniques et Sciences Informatiques*, 19, 1223–1244.
- [39].Luong, N. C., et al. (2019). Deep reinforcement learning in communications. *IEEE Communications Surveys & Tutorials*, 21(4), 3133–3174.
- [40].Shah, S. A. A., Ahmed, E., Imran, M., & Zeadally, S. (2018). 5G for vehicular communications. *IEEE Communications Magazine*, 56(1), 111–117.
- [41].Precedence Research. (2021). Internet of Medical Things (IoMT) market growth. <https://www.precedenceresearch.com>
- [42].Aman, A. H. M., et al. (2021). IoMT amid COVID-19 pandemic. *Journal of Network and Computer Applications*, 174, 102886.
- [43].Chen, M., et al. (2023). Edge intelligence in 6G: Architecture, key technologies, and applications. *IEEE Network*.
- [44].Liu, Y., et al. (2024). 6G-enabled edge intelligence for smart applications. *IEEE Internet of Things Journal*.
- [45].Zhang, J., et al. (2022). AI-driven edge computing for smart cities and Industry 4.0. *Future Generation Computer Systems*.
- [46].Wang, T., et al. (2023). Edge AI for next-generation IoT systems. *IEEE Transactions on Industrial Informatics*.
- [47].Chen, M., et al. (2023). Edge intelligence in 6G. *IEEE Network*.
- [48].Liu, Y., et al. (2024). 6G-enabled AI networks. *IEEE Internet of Things Journal*.
- [49].Zhang, J., et al. (2022). AI-driven edge computing for smart networks. *Future Generation Computer Systems*.
- [50].Wang, T., et al. (2023). Edge AI for next-generation wireless systems. *IEEE Transactions on Industrial Informatics*.